

THE UNIVERSITY OF LAHORE

Department of Computer Science & IT

Faculty of Information Technology

Final Year Project Proposal

	Day		Month			Year			
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PRODUCT TITLE:	Gym Trainer Model
KEY WORDS:	Calculate Gym Exercise Repetitions - Correct Posture While Exercising - Detects Body Angles - Error Detection while doing wrong Exercise
DOMAIN OF THE PRODUCT:	Health & Fitness Technology / Human Pose Estimation / Computer Vision in Exercise Monitoring
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1- Problem Identification:

Most people who exercise at home or in commercial gyms without expert supervision tend to do exercises with poor posture, resulting in poor workouts or worse, serious injury. Personal trainers are costly, not always available, and can't observe several clients at a time. The demand for cheap, real-time feedback tools that can assist fitness enthusiasts to stay in correct form, monitor progress, and avoid injuries is increasing, particularly with the remote and home-based exercise trend.

Although fitness apps and video tutorials gained popularity, neither of them presents real-time correction of posture along with rep count based on computer vision in any way that truly imitates supervision by a person.

Your project seeks to plug this gap with the development of a smart, ML-powered gym trainer that prevents users from performing exercises incorrectly or injuriously, integrating pose detection, angle measurement, and repetition counting — something that is not well served in the mainstream fitness technology marketplace.

1- Product Market Analysis:

The global fitness tech industry is exponentially growing due to health-conscious growth and growing trends for at-home workout culture, particularly after the pandemic. Fitness enthusiasts, non-trainees lacking trainers for their work out routines, post-recovery patients, and far-distant or work-from-home exercisers make up the basic targeted demographics of this venture. Leading players such as Fitbit, Peloton, and Nike Training Club offer overall fitness tracking or guided exercise but not real-time posture correction through computer vision. Software that tries pose estimation, like Freeletics or Kaia Health, tends to be limited in precision or require manual input. This suggests a significant market potential for an AI-powered virtual trainer that provides automated form correction, repetition tracking, and real-time feedback. Potential clients are individual users, gyms, fitness startups, and rehab centers looking to improve training efficiency and safety at economical, scalable costs.

2- Unique Value Proposition:

Real-time Posture Detection and Correction: Unlike most fitness apps that provide pre-recorded guidance, this system uses computer vision to analyze body posture live and immediately flag incorrect form, helping users avoid injuries.

Accurate Rep Counting Based on Movement and Angles: The model doesn't rely on wearable sensors or manual inputs but instead uses pose estimation and angle calculation to count reps automatically and accurately.

Body Angle Detection for Form Assessment: By calculating joint angles (e.g., elbow, knee, shoulder), the system can evaluate form with high precision, enabling customization form-checking logic for different exercises.

Hardware-Free and Camera-Based Solution: This solution only needs a camera (webcam or smartphone) and does not require expensive wearables, making it cost-effective and easy to use.

3- Features:

Real-Time Posture Detection:

Uses computer vision (e.g., OpenPose, Mediapipe) to continuously monitor the user's body posture during exercise.

Incorrect Form Detection and Alerts:

Identify when the user is doing an exercise incorrectly according to body angles and posture deviation, and provide visual or audio feedback instantaneously to correct it.

Automated Rep Counting:

Automatically counts exercise repetitions by monitoring body movement and angle change, minimizing manual tracking.

Body Angle Measurement:

Computes essential joint angles (e.g., knees, elbows, shoulders) to determine the quality of exercise form and ensure every repetition satisfies the proper range of motion.

Multi-Exercise Support:

Provides support for numerous typical exercises such as squats, push-ups, lunges, and planks with dedicated detection logic for each.

User-Friendly Interface:

A minimalist and easy-to-use interface offering real-time feedback, visual cues, and progress tracking to enhance user interaction.

No Wearables or Additional Hardware Needed

Works exclusively off a standard webcam or smartphone cam, making it affordable and universally accessible.

Performance Feedback and Summary:

Provides after-every-session summary of reps, accuracy level, and trouble spots for continuous tracking of improvement.

4- Revenue Generation:

Subscription-Based Model:

Provide users with a freemium model with fundamental features (restricted exercises and feedback), but charge on a monthly or yearly subscription for advanced features like complete exercise library, custom plans, detailed progress updates, and in-depth performance analysis.

In-App Advertising:

Show non-obtrusive, relevant ads in the free version of the application, especially by sports brands, nutrition supplements, or sportswear firms, earning ad revenue while maintaining core services for free for basic consumers.

Partnerships with Fitness Studios and Gyms:

License the technology to fitness centers, personal trainers, and gyms that can use it as part of their client programs as a virtual assistant, enabling them to scale their services and enhance member engagement.

Corporate Wellness Programs:

Partner with corporate wellness platforms to provide the app as part of employee fitness programs. Businesses are increasingly investing in digital wellness solutions to keep their employees active and healthy.

One-Time Purchase or Lifetime Access Plan:

Offer users the ability to purchase premium features once for lifetime access, targeting users who do not like subscriptions.

Affiliate Marketing:

Promote fitness gear or accessories within the app (e.g., yoga mat, dumbbell) and receive affiliate commission from programs with online stores like Amazon or Decathlon.

Custom White-Label Solutions:

Create a white-label version of the app that will be branded and sold to personal trainers or fitness influencers who wish to introduce their own branded virtual training platform.

5- Marketing and Sales Strategy:**Target Market & Customer Segments:**

- Home fitness customers seeking economical and guided workouts.
- Novice gym members requiring posture adjustment and repetition guidance.
- Fitness coaches and personal trainers looking for digital support in clients.
- Health clubs and fitness studios wanting to use AI-equipped smart mirrors or self-guidance training.
- Physical therapy facilities for posture correction and movement tracking within rehabilitation programs.
- Corporate wellness programs to enhance employees' fitness.

Marketing Channels:Social Media Marketing:

Utilize channels such as Instagram, YouTube, and TikTok to exhibit demo videos, before/after changes, consumer testimonials, and form correction demonstrations.

Influencer Partnerships:

Participate in deals with fitness experts and YouTube personal trainers to introduce the tool and illustrate its merits to their listeners.

Content Marketing (SEO):

Release fitness websites, workout tricks, and protection against injuries booklets to ensure organic visits and establish the firm as a resourceful expert.

App Stores (if mobile-based):

Maximize app listing with compelling images, concise descriptions, and favorable reviews for Google Play and Apple App Store visibility.

Fitness Communities & Forums:

Advertise the product on sites such as Reddit's r/fitness, Facebook groups focused on fitness, and workout forums online.

Email Campaigns:

Make use of email newsletters to reactivate users, report progress summary, provide tips, and advertise new features or premium subscription.

Customer Acquisition Strategies:

Free Trial:

Provide a 7–14 day free trial or free basic model to entice users and upgrade them to premium with in-app purchases.

Referral Program:

Reward friends invited by users with discounts or premium days for free.

Early Access Beta Testing:

Grant early access to the product in return for feedback and testimonials, building community loyalty and word-of-mouth promotion.

Bundle with Fitness Equipment or Subscriptions:

Collaborate with fitness equipment vendors to bundle the app with purchases of equipment (e.g., resistance bands or yoga mats).

Pricing Models:

Free Model:

Free with basic features (e.g., single exercise type, limited feedback). Full features are unlocked with paid plans.

Subscription Plans:

Monthly: \$9.99/month

Annual: \$79.99/year (premium)

Team/Trainer Plan: Special rate for coaches or fitness centers

Lifetime Plan:

One-time fee (e.g., \$149) for complete, unlimited access — best for long-term members.

6- Risk Assessment:

The creation of an AI-powered gym trainer offers a number of possible risks and challenges in technical, market, and financial spaces. Technically, the accuracy of pose detection can degrade in low-light or messy conditions, and real-time processing can be slow on low-end hardware; these can be addressed through the use of optimized models such as Mediapipe and providing lightweight or server-side solutions. Market risks are user resistance to AI-powered fitness products and competition from well-established fitness apps, and these can be managed through differentiation in the form of real-time correction without the need for hardware and effective user engagement strategies. Financial risks are high development expense and uncertain monetization, but these are minimized by prioritizing an MVP focus, freemium models, and cloud-based development tools. Moreover, data privacy issues associated with camera-based tracking can be addressed by clear privacy policies and local processing capabilities. Anticipating these issues is key to upholding the product's technical robustness, user confidence, and long-term viability.

7- Scalability and Growth Potential:

The scalability and growth potential of the AI-powered gym trainer project are very high because of some very important reasons. Firstly, market demand for fitness technology keeps increasing as a result of a growing global interest in health and wellness, home workouts, and digital fitness equipment, particularly post-pandemic. As more individuals look for convenient, low-cost, and personalized fitness solutions, the market for virtual trainers and exercise monitoring systems is likely to grow at a rapid pace.

The scalability of technology is assisted by the versatility of AI and machine learning models, which have the ability to be applied to various exercises and different levels of fitness. The application of computer vision libraries such as Mediapipe and OpenCV makes it simple to adapt to new exercises, while cloud computing and mobile optimization guarantee that the technology can be implemented to a wide range of people, from low-end mobile users to high-performance device users. Moreover, the fact that the system is based on cameras instead of wearables lowers the entry barrier, making it accessible to everyone.

Market opportunities for expansion consist of providing personalized solutions for health clubs, private trainers, and corporate wellness, where the product can be licensed as a vehicle for client activation and fitness tracking. Additional growth would be facilitated through collaborations with health equipment brands, corporate wellness software, or penetrating the post-injury recovery and rehabilitation/physio market. Moreover, international scaling is possible because fitness is a universal activity and the solution is low-cost and hardware-free, thereby offering appeal in global, various markets.

In summary, the project has high growth potential in terms of both user acquisition and technical scalability, with potential for expansion into new markets, partnerships, and international growth.

8- Implementation and Execution Plan:

1. Project Overview and Objective:

The goal of this project is to develop an AI-based gym trainer that can detect body posture, provide real-time feedback on incorrect form, count reps, measure body angles, and support multiple exercises. This will be implemented using Python libraries such as Pandas, Numpy, Mediapipe, OpenCV, and Matplotlib.

2. Project Phases & Milestones:

Phase 1: Planning and Research (1-2 Weeks)

Milestone 1: Market Research & Finalization of Features

- Competitor research and similar tools.
- Finalize MVP features to be included.
- Establish key metrics for posture detection, rep counting, and form correction.
- Distribute team members to respective tasks and responsibilities.

Phase 2: Data Collection and Preparation (2-3 Weeks)

Milestone 2: Dataset Collection and Labeling

- Collect sample video data or utilize pre-labeled data for human pose estimation (e.g., COCO, MPII).

- Develop labeled datasets to validate the rep counting and form detection algorithm.
- Develop synthetic datasets in case real-world data is not available.
- Data preprocessing and cleaning (using Pandas for data manipulation).

Team Assignment:

Member 1: Research and acquire datasets, pre-process data for pose detection.

Member 2: Handle data preprocessing (data manipulation, cleaning).

Member 3: Research alternative datasets and make sure they are compatible with the model.

Phase 3: Algorithm Development and Integration (3-4 Weeks)

Milestone 3: Pose Detection Model Development

- Implement and test Mediapipe or OpenPose for real-time pose detection.
- Integrate OpenCV for video processing and frame extraction.
- Implement rep counting through body movement and angle changes (using Numpy for numerical computation).
- Compute body angles (shoulder, knee, elbow) to determine proper form using Numpy and Matplotlib.

Team Assignment:

Member 1: Pose estimation, integrating with OpenCV for real-time video processing.

Member 2: Implementation of the rep counting algorithm and body angle measurement.

Member 3: Error detection and feedback system (create alerts).

Phase 4: User Interface and Feedback Mechanism (2-3 Weeks)

Milestone 4: User Interface Development

- Create a simple user interface for visual feedback (form feedback, reps, angles).
- Utilize Matplotlib or GUI libraries (such as Tkinter or PyQt) for presenting feedback.
- Enforce real-time visual indicators of correct/incorrect form.
- Create feedback mechanisms (alerts, color-coding to correct posture).

Team Allocation:

Member 1: Develop integrating the pose detection with the interface.

Member 2: Handle the feedback and alert system.

Member 3: Create and implement the front-end user interface.

Phase 5: Testing and Refinement (2 Weeks)

Milestone 5: System Testing and Bug Fixing

- Test the application on different devices (webcams, smartphones).

- Detect bugs in pose estimation, rep counting, and body angle detection.
- Correct bugs and improve performance.
- Gather user feedback and refine the design.

Team Assignment:

Member 1: Perform system testing on pose detection and generate alerts.

Member 2: Perform rep counting and accuracy testing

Member 3: Improve performance and debug the UI.

Phase 6: Final Presentation and Documentation (1-2 Weeks)

Milestone 6: Documentation and Final Demo

- Develop a detailed report/document that outlines the workflow of the project, technologies employed, challenges encountered, and outcomes.
- Capture a demo video of the working prototype.
- Create a final presentation (Google Slides or PowerPoint) for project submission.

Team Assignment:

Member 1: Technical documentation (pose detection model, algorithms).

Member 2: Testing outcomes, challenges, and optimization procedures.

Member 3: Final presentation development and demo video.

3. Resource Management:

Software/Tools:

Python Libraries: Matplotlib, Pandas, OpenCV, Numpy, Mediapipe

IDE/Editor: Visual Studio Code, PyCharm

Database/Storage: Local database or cloud-based storage for dataset

Testing Equipment: Multiple webcams/smartphones for testing

Hardware:

Laptops/computers with adequate CPU/GPU for testing and training of models.

Webcams or smartphones for video in during testing.

Supervisor's Signature:

Suggested Improvements regarding FYP (*if any*):
